Lab 2: Feedback Control of a Ping Pong Ball

Christian Landrum & Wyatt Probst

Exercise 1

1.1: Setup Sonar Sensor

Exercise 1 was setting up the Sonar Sensor to read and display distances to an object

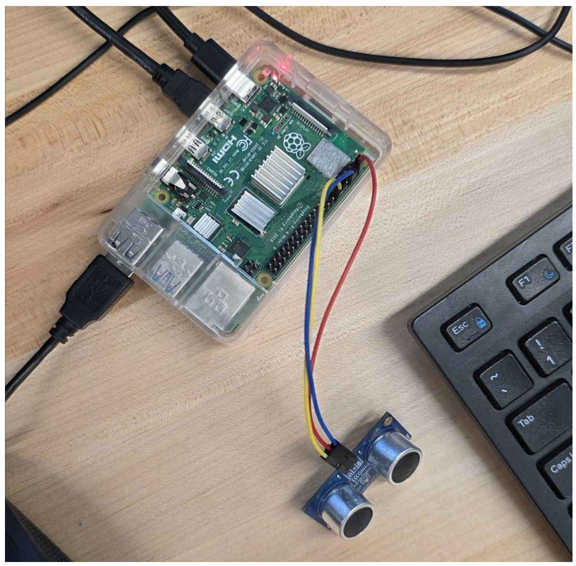

Figure 1: Sonar Sensor Connection

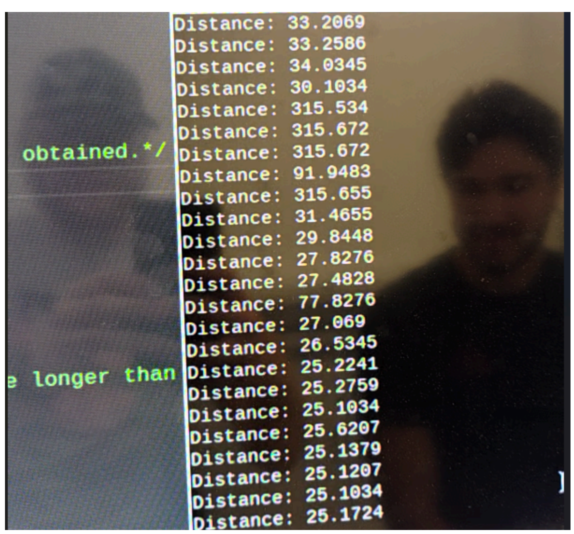

Figure 2: Sonar Sensor Output

Exercise 2

2.1 Ping Pong Ball PID

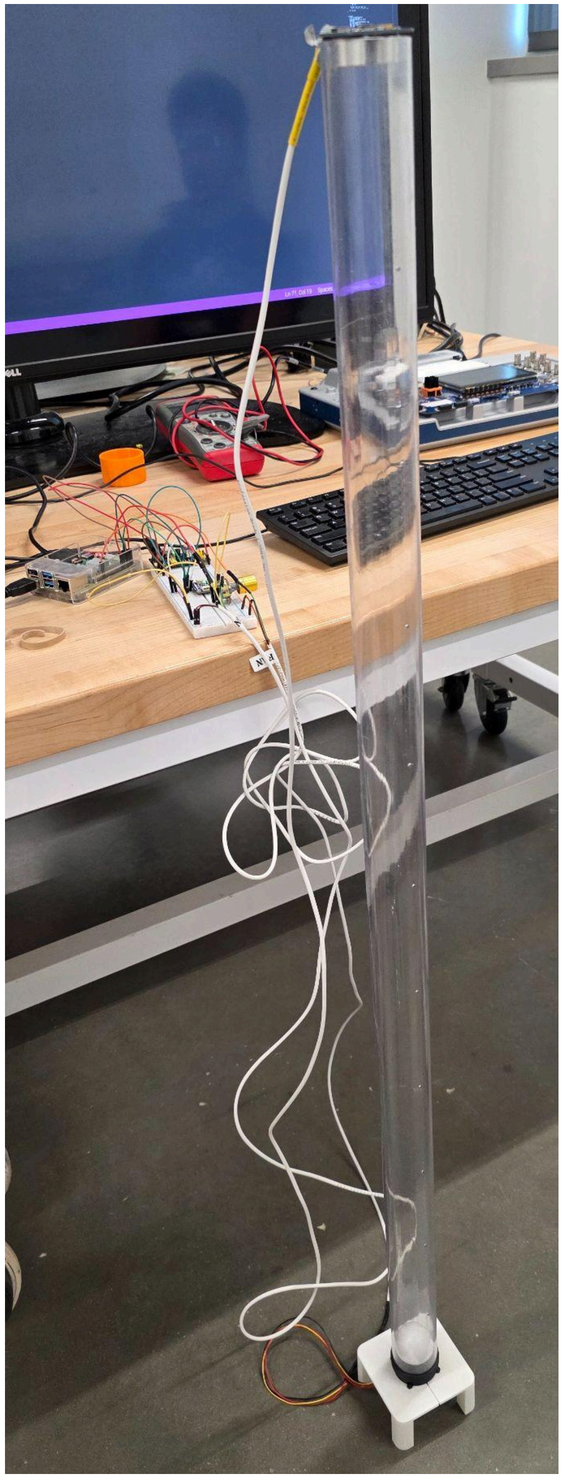

Figure 3: PID Loop connection

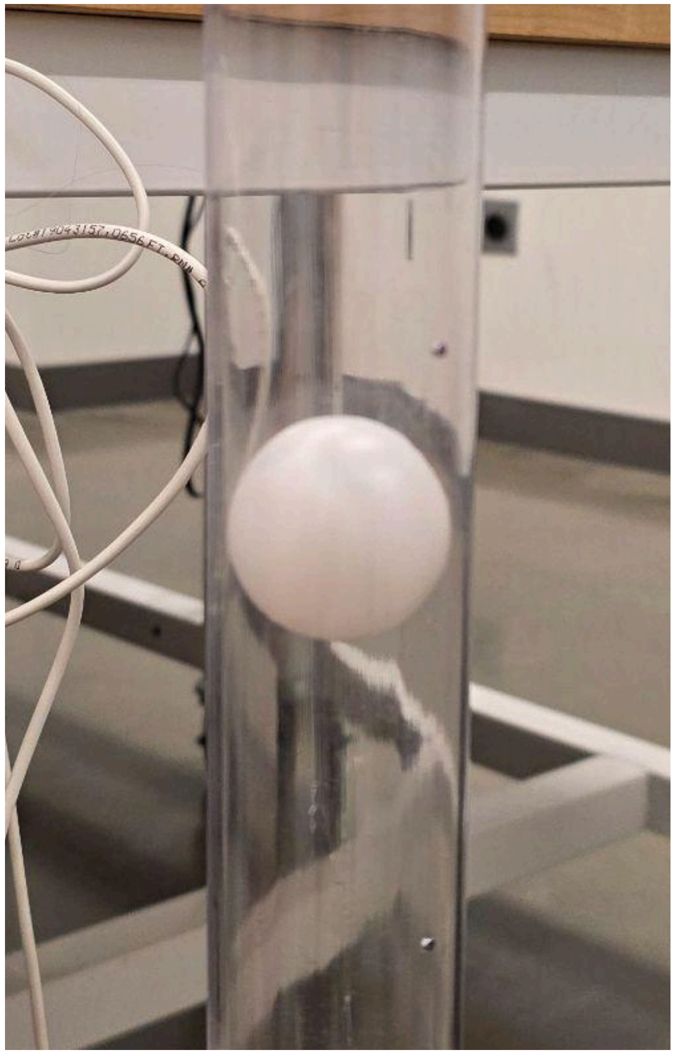

Figure 4: Ping Pong ball hovering using PID Loop

Supplemental Questions

- **Briefly summarize what you learned from this lab.**

  In this lab we learned how to use a sonar sensor with the GPIO on the Raspberry Pi. We also learned about how to integrate automatic feedback into our program and circuit using the P, I, and D parameters.

- **In PID control, how will the values of the P, I, and D parameters affect your control performance?**

  Increasing P decreases rise time, increases overshoot, and decreases steady state error. Increasing I also decreases rise time and increases overshoot, but increases settling time. Increasing D decreases overshoot and settling time.

- **How did you decide the value of the P, I, and D parameters to achieve a good control performance?**

  We achieved the values based on watching how the ping ball reacted to changes based on what parameter we changed. If the ball would oscillate after a change, we would decrease either P or I to remove it and then alter to get a responsive rise time and settle time while staying at the set height.

ACKNOWLEDGMENTS

I certify that this report is my/our own work, based on my/our personal study and/or research and that I/we have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication. I/We also certify that this assignment/report has not previously been submitted for assessment anywhere, except where specific permission has been granted from the coordinators involved.

Author-1 Signature **Christian Landrum**
Author-2 Signature **Wyatt Probst**

REFERENCES

1) Lab 2 Slides

2) Lab 2 Handout

3) Lab 2 Documents